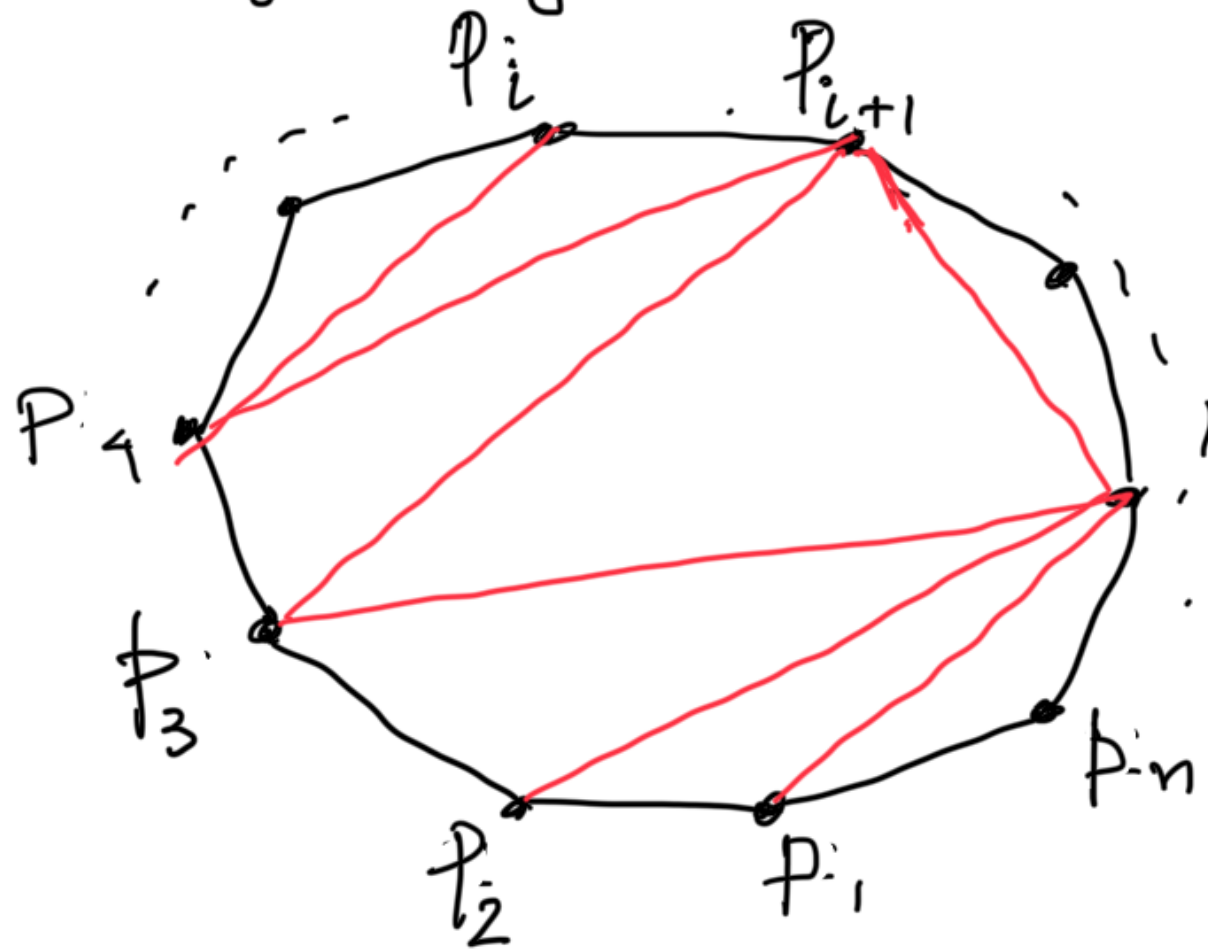
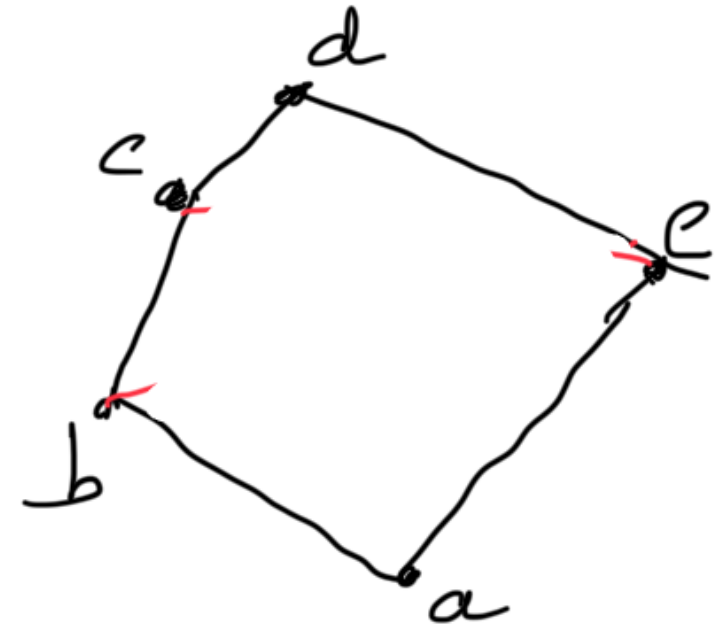
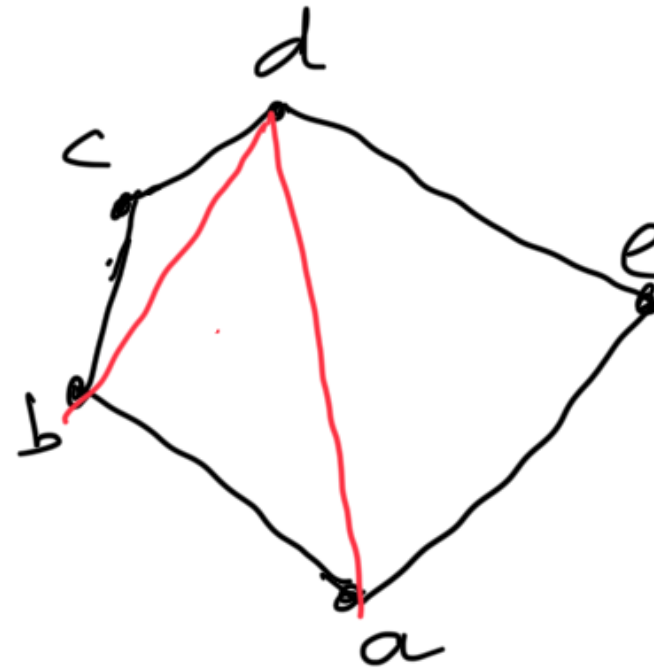
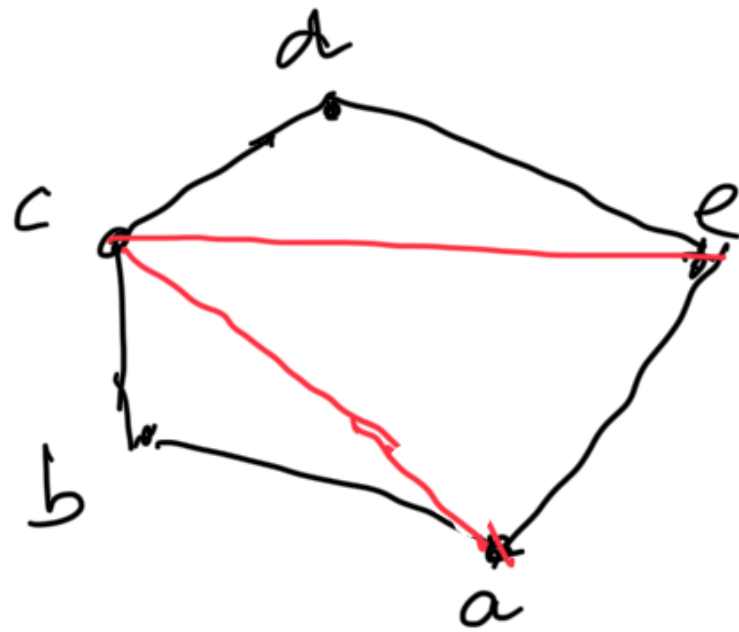
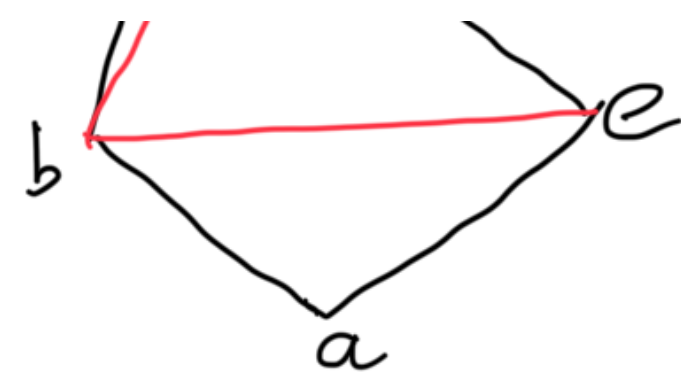
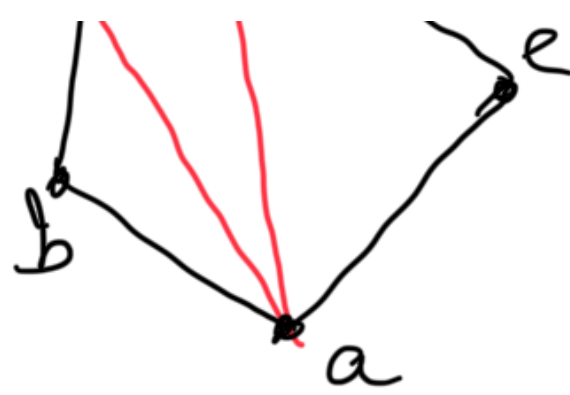
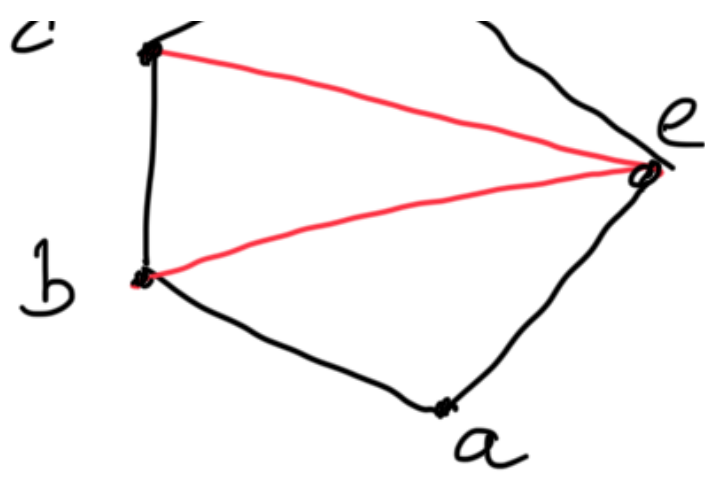


Triangulating a Convex n -gon

$$p_i = (x_i, y_i)$$

non-intersecting $n-3$ diagonals
(among a possible $\binom{n}{2} - n$ diagonals)



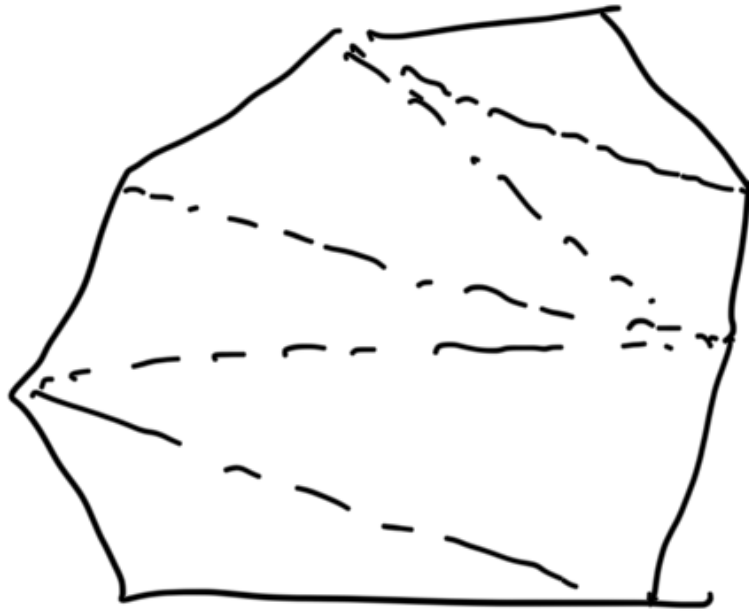


Which is best?

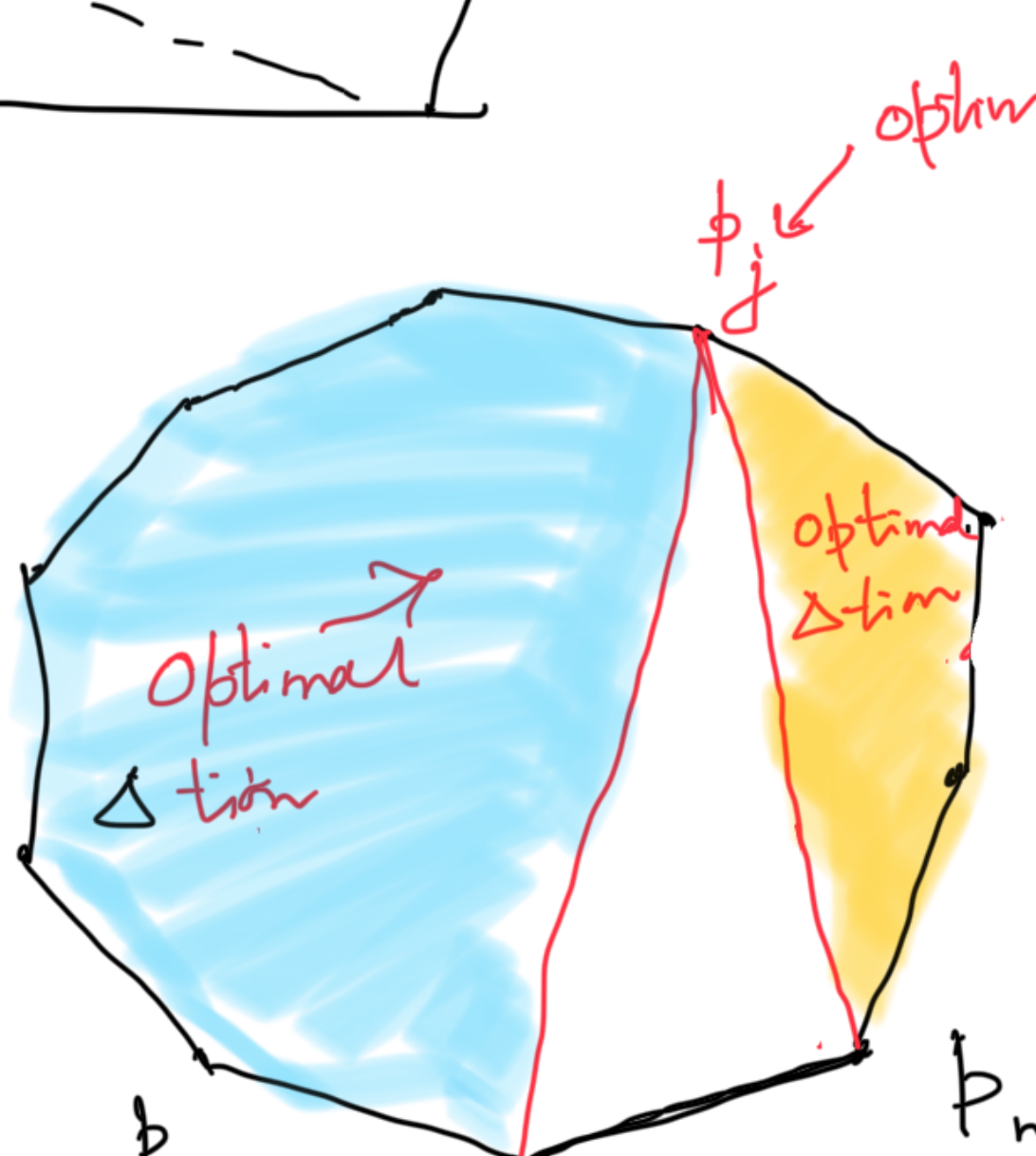
Minimum weight Δ -tion

Greedy Δ -lin : Choose the diagonal
of smallest length that does

not intersect the previous diagonals



Does greedy lead
to min wt ??



optimal point

Property of
optimal Δ -tion

Let $C(s, t)$
denote the optimal
cost of Δ -tion

p_i for subpolygon $(s, s+1, \dots, t)$

$$C(p_s, p_e) = \min_{s \leq j \leq e} \left\{ C(p_s, p_j) + C(p_j, p_e) + l(p_s, p_j) + l(p_j, p_e) \right\}$$

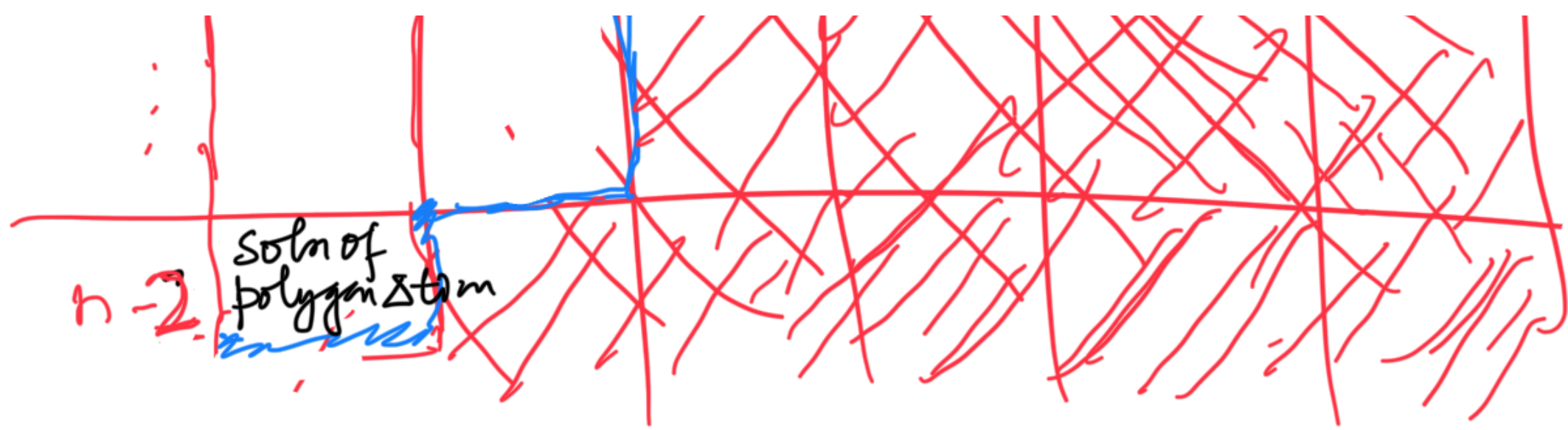
$e > s$

Base case when (p_s, p_{s+1}, p_e) is a Δ and the cost = 0

$s=1$	1	2	3	$n-2$
$e-s=1$	0	0		cost = 0 since no Δ can be formed by 1 edge
2	$l(s, e)$	length of the edge (s, e) in the $\Delta(s, s+1, e)$		
3				
$\vdots$				
i				

$\uparrow$ # vertices in the subpolygon

as per recurrence: min of i pairs



Obs : To fill an entry in row i
(corresponds to $s-e$), we have to consult about
 $i-1$ entries of the previous rows (which have been
computed previously)

Running time analysis :

Each entry in row i take $O(i)$ time

$\Rightarrow$ $(n-i)$ entries in row i take
 $O((n-i)i)$ time

$$\text{Total time} = \sum_{i=1}^{n-2} (n-i)i$$

$$\approx \sum_{i=1}^n n \cdot i = n \sum_{i=1}^n i$$

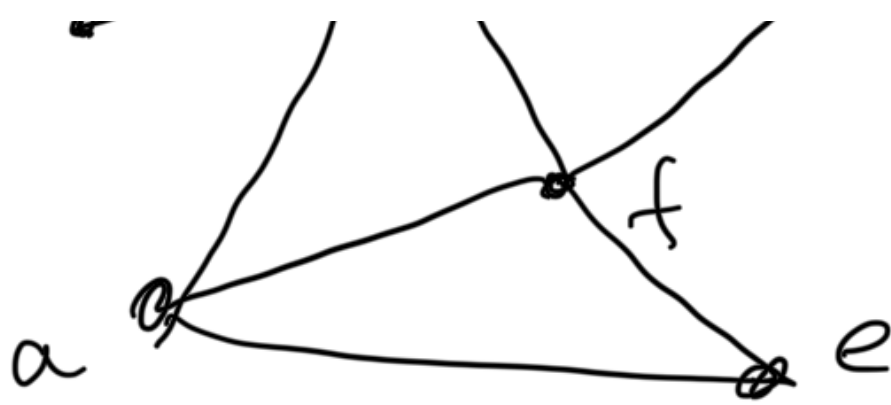
$$= O(n^3)$$

Finding independent subsets of vertices

$$G = (V, E)$$



A subset $S \subseteq V$ is



independent if

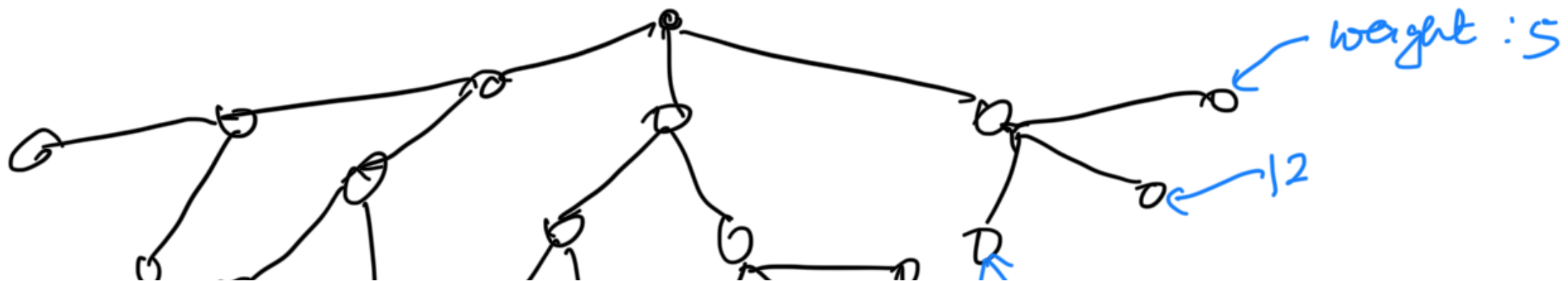
$$\forall x, y \in S$$

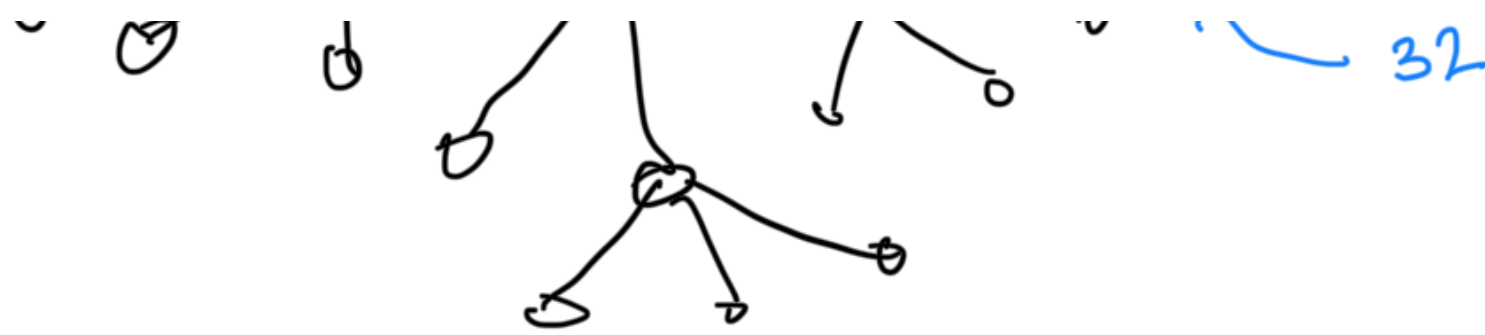
$$(x, y) \notin E$$

Find the largest independent subset of V

Problem has no known polynomial time solution for general graphs

How about a tree?





Ex Set of leaf nodes are independent

Weighted : If we have weights on each vertex
independent set
find the maximum weighted independent set of node